Strategies for Improved dCO_2 Removal in Large-Scale Fed-Batch Cultures

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Carbon dioxide buildup in large-scale reactors can be detrimental to cell growth and productivity. In case of protein X, a therapeutic glycoprotein, when cultures were scaled up from bench scale to the pilot plant, there was a 40% loss of specific productivity. The dissolved CO_2 (dCO_2) level was 179 ± 9 mmHg at the pilot plant scale and 68 ± 13 mmHg at bench scale. The authors proposed a comprehensive approach to maintain dCO_2 levels between 40 and 120 mmHg throughout the 14-day fed-batch process. A cell-free experiment was used to investigate the impact of the following parameters on dCO_2 removal: (1) sparge rate, (2) agitator speed, (3) bubble size, (4) bicarbonate concentration, (5) impeller position, and (6) aeration rate at the headspace of bioreactor. dCO_2 was measured using a fiber optic based probe. dCO_2 removal rate was a strong function of sparge rate and a weak function of agitator speed. Bubble size was modulated by the presence or absence of a sparge stone (10 μm pore size, 1 cm pipe i.d.). Open pipe provided 3- to 4-fold better dCO_2 removal for the same mass transfer coefficient (k_La) value. A mathematical model and a bench-scale experiment indicated that the benefit of a lower level of sodium bicarbonate in the culture medium was transient for batch and fed-batch cultures. Thus, this strategy was not used at pilot scale. Decreasing top impeller position improved k_La of dCO_2 by 2-fold. Changing headspace aeration rate from 0.02 to 0.04 vvm had no impact on dCO_2 removal. Two pilot runs were conducted using (A) open pipe and (B) antifoam in the presence of sparge stone, both in conjunction with lower impeller position. The presence of antifoam may interfere in product purification; however, demonstration of antifoam removal can be difficult. Open pipe allowed an alternative to using antifoam, as foam level with open pipe was significantly less. Both strategies successfully reduced dCO_2 level by 2.5-fold (179 ± 9 vs 72 ± 9 mmHg). Titer at day 10 of culture improved by 1.5-fold. Specific productivity improved by 41%. Historically, cultures were harvested around day 9−11 because of the high amount of foam; both strategies allowed the cultures to be extended up to day 14, resulting in 2-fold higher titer compared to that of the historical control without compromising protein quality.

Introduction

One of the recurrent issues in industrial mammalian cell culture has been accumulation of dissolved CO_2 (dCO_2) in large-scale bioreactors. In small-scale reactors a significant portion of dCO_2 stripping occurs through surface aeration. In large-scale reactors, the liquid surface-to-volume ratio decreases and overall dCO_2 removal becomes less effective.

Several groups have studied the impact of dCO_2 on cell culture (*1*−*9*). For CHO cells expressing recombinant human macrophage colony stimulating factor, cell growth rate and specific productivity decreased to 52% and 56% of control, respectively, when dCO_2 level increased from 53 to 165 mmHg (*3*). For NS/0 cells, a change of dCO_2 from 60 to 120 mmHg led to 20% decrease in IgG yield; however, specific productivity was not affected (*1*). In a CHO perfusion culture producing a viral antigen, increase in dCO_2 from 35 to 148 mmHg led to decrease in cell density and specific productivity of 33% and 44%, respectively (*5*). For hybridoma cells, a 63% decrease in growth rate was observed at dCO_2 level of 195 mmHg; however, specific productivity was unchanged (*2*). For perfusion cultures of BHK cells, at 300 mmHg dCO_2, specific productivity decreased by 40% compared to control. High dCO_2 also led to decreased glucose, lactate, and glutamine specific metabolite rates (*7*). The inhibitory effects are at least in part due to the effects of dCO_2 on metabolic pathways. Increased dCO_2 causes perturbation in intracellular pH, thereby affecting pH-dependent enzymatic reactions (*7*, *9*, *10*). For example, phosphofructokinase, which is a rate-limiting enzyme in the glycolytic pathway, is inhibited by low intracellular pH (*11*). Excessive stripping of dCO_2 is also detrimental to cell growth (*7*, *12*), which suggests that there is likely an optimal level of dCO_2 for cell culture. This optimal value would most likely vary from cell line to cell line.

During scale-up of a CHO cell line based fed-batch process from bench scale to 1000 L, we observed significant increase in dCO_2 level. In this paper we designate the product, a therapeutic glycoprotein, as protein X. The end of culture dCO_2 level at the 1000-L scale was 179 ± 9 mmHg and at 1.5-L scale was 68 ± 13 mmHg. This paper describes our road map to find an easily imple-

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Table 1. 1000-L Cell-Free Experiment Design

stone/ pipe	s	s	s	s	s	s	s	s	p	p	p	p	p	p	p	p
top impeller (high/low)	h	h	h	h	l	l	l	l	l	l	l	l	l	l	l	l
sparge (vvm)	0.002	0.006	0.006	0.01	0.006	0.006	0.006	0.01	0.006	0.006	0.006	0.01	0.01	0.01	0.01	0.01
tipspeed (m/s)	1.3	1.3	1.8	1.8	1.3	1.3	1.8	1.8	1.3	1.3	1.8	1.8	1.8	2.6	3.5	1.3
overlay (vvm)	0.02	0.02	0.02	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
antifoam (y/n)	n	y	y	y	n	y	y	y	n	y	n	n	y	n	n	n

mentable, GMP (good manufacturing practice)-amenable method to maintain low dCO_2 level at 1000-L scale. Although many publications have used mathematical models and small-scale studies simulating large-scale dCO_2 levels to describe dCO_2 control strategies, at scale data is relatively rare. In this paper, we took a comprehensive look at possible strategies and conducted a side-by-side comparison of two 1000-L runs using the two final strategies. We also generated a mathematical model to delineate the contribution of sodium bicarbonate in the fed-batch culture dCO_2 level.

Materials and Methods

Cell-Free Study at 1.5-L Scale. The purpose of this experiment was to determine the impact of the following parameters on dCO_2 removal: (1) sparge rate, (2) agitator speed, and (3) bubble size (*5*, *13*). Two 1.5-L reactors, one with a 15-μm sparger and another without a sparger (open pipe, 1 cm i.d.) were filled with in-house medium. A pH probe, a DO probe, and a dCO_2 probe (YSI Biovision 8500) (*14*) were inserted in each reactor. Nitrogen was sparged into the reactors until the DO level reached zero and then CO_2 was sparged into the reactors until dCO_2 level was around 140 mmHg. At this stage a 1:1 ratio of air and oxygen was sparged into the reactors until dCO_2 level decreased to 100 mmHg. The 1:1 ratio of air and O_2 was used to emulate the final proportion of air to O_2 in the reactor at the end of culture.

Cell-Free Study at 1000-L Scale. A cell-free experiment at 1000-L scale was conducted to investigate the impact of the following parameters on dCO_2 removal: (1) sparge rate, (2) agitator speed, (3) bubble size, (4) top impeller position, (5) headspace aeration rate, and (6) antifoam (Table 1). Two 1000-L reactors were used for this study. The sparge stone of one of the bioreactors was removed and the top impeller position for both of the reactors were altered during the 4-day study. The reactors were sterilized and filled with in-house medium, N_2 was sparged into the reactor to bring DO to zero, CO_2 was sparged until the dCO_2 level reached ~140 mmHg, and then a 1:1 mixture of air and oxygen was sparged into the tanks until dCO_2 levels reached 100 mmHg. pH was controlled at 6.9 by NaOH addition up to the start of air/O_2 sparging.

Small-Scale Cell Culture Study on Sodium Bicarbonate Concentration. Two identical CHO cultures were conducted at 1.5-L scale, at 37 °C in the in-house medium with 2 and 0.5 g/L of sodium bicarbonate. The sparging protocol (vvm of air and oxygen) was similar for the 1.5-L reactor runs and the historical 1000-L reactor runs. Each reactor contained only one impeller. pH was maintained at 6.9 by adding CO_2 at the earlier part of cultures and by adding NaOH at the later part of cultures. Tip speed was maintained at 0.42 m/s, and the DO was maintained at the desired setpoint by ramping up air sparge and supplementing with O_2 sparge.

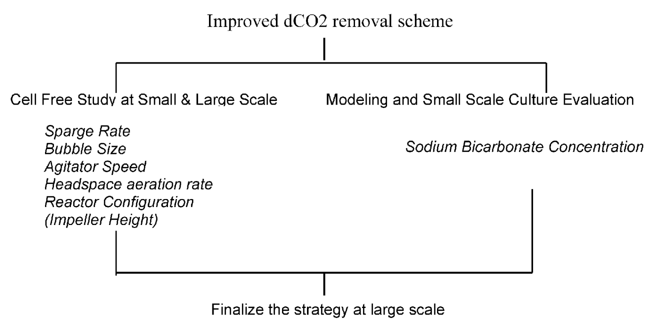
Proof of Concept Runs at 1000-L Scale. The 1000-L reactors used two impellers and either a sintered 10-μm sparge stone or an open pipe (1 cm i.d.). CHO cultures were grown in the in-house medium at 37 °C with a bicarbonate concentration of 2 g/L. pH was maintained at 6.9 using CO_2 gas and a NaOH feed. Tipspeed was maintained at 1.3 and 2.3 m/s with sparger and open pipe, respectively. DO was maintained at set point by ramping up air sparge and supplementing with O_2 sparge. When a sparger was used, a silicone emulsion based antifoam was used at a concentration of 100 ppm.

Results and Discussion

Historical Data. We have observed that accumulation of high level of dCO_2 in the culture broth caused significant loss of productivity at the pilot plant 1000-L fed-batch cultures. In case of protein X historical runs, dCO_2 level at the time of harvest was 179 ± 9 mmHg, a figure that was substantially higher than the 68 ± 13 mmHg dCO_2 level observed at harvest for 4-L lab-scale reactors. As Figure 1 indicates, dCO_2 accumulation can be an issue related to scale-up. As a result of excessive foaming, the vvm of air sparged for the 1000-L GLP reactors from day 7 through 14 of culture was less than the vvm of air used at smaller scale (0.0015 vs 0.00175 vvm), which also contributed to dCO_2 accumulation at the 1000-L scale. The net impact of elevated dCO_2 was a 40% decrease of specific productivity.

dCO_2 Control Strategies. A comprehensive evaluation of possible strategies for improved dCO_2 removal for fed-batch culture was conducted and the final strategies were successfully implemented in 1000-L cell cultures.

Following is a schematic of the different strategies that were considered: Increased sparge rate and agitator



speed leads to increased k_La. Typically, k_La is considered a strong function of sparge rate and a relatively weak function of agitator speed. For similar values of k_La, larger bubbles have been found to provide better dCO_2 removal (*5*, *13*). Headspace aeration rate improves surface gas exchange through improving surface k_La. Sodium bicarbonate is the most commonly used buffer in culture media, which is also a source of dCO_2. In case of CHO perfusion cultures, reduction of sodium bicarbonate has been shown to reduce dissolved dCO_2 level (*5*). We looked at reduction of sodium bicarbonate for fed-batch cultures as a possible means of reducing dCO_2. Reactor configuration (e.g., number and position of impeller(s), baffles, sparger) significantly impacts the mixing condition in the reactor. Some typical design values for bioreactors are (*15*) H(btm) = D(i), H(top) = D(i), H(i) = 1 to 1.5 × D(i) (Figure 2)

Prior studies in these 1000-L reactors had indicated that the position of the sparger and that of the bottom

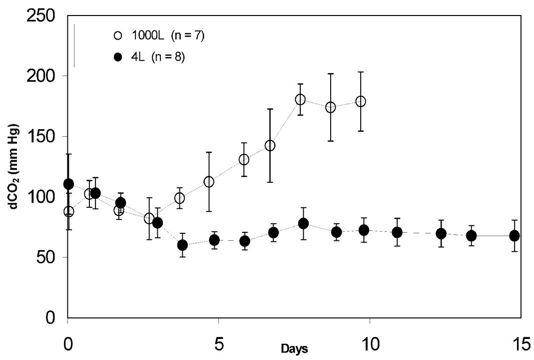


Figure 1. dCO_2 in different scale reactors.

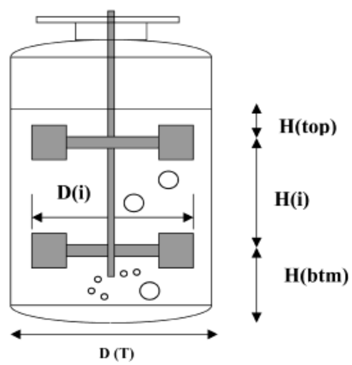


Figure 2. 1000-L reactor configuration.

Table 2. Impact of Sparge Method on k_La at 1.5-L Scale

	k_La (h^{-1})			
sparge scheme, tipspeed	0.0067 vvm	0.0133 vvm	0.033 vvm	0.067 vvm
sparger, 0.42 m/s	14.88	19.40	118.33	188.07
open pipe, 0.42 m/s	0.80	1.06	2.64	3.38
open pipe, 0.63 m/s	1.03	1.46	3.55	7.4

impeller had relatively weak impact on mixing (data not shown). For this study only the impact of the height of the top impeller was investigated at 1000-L scale.

1.5-L Cell-free Study. As indicated by Figure 3a and b, the difference between dCO_2 removal rates with a sparge stone and an open pipe depends on the vvm of the sparged gas. For a given vvm, dCO_2 removal rate with an open pipe was 28−76% of that with a sparger for the same agitation rate and was 36−134% of that with a sparger for increased agitation rate. However, as Table 2 indicates, for open pipe the k_La value for a given vvm was only 2−7% of the k_La value for a sparger. The large bubbles created by open pipe resulted in reduction of total gas−liquid interfacial area, which would explain the reduced k_La values for open pipe with a given vvm. These data indicate that dCO_2 removal rate for a given vvm depended not on k_La exclusively. The higher driving force with larger bubbles partially counteracted the impact of lower k_La for open pipe. Visual observation indicated that increased agitation increased residence time of bubbles, which possibly improved dCO_2 removal rate. Increased agitation may have reduced bubble size, which then

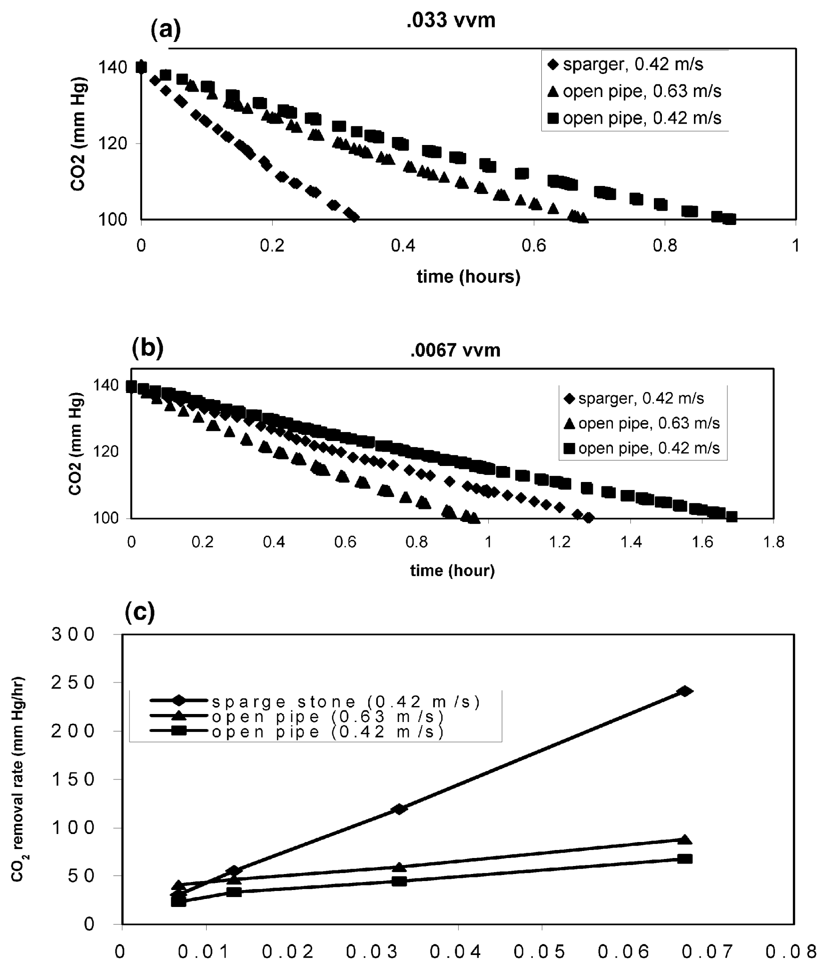


Figure 3. Data from 1.5-L reactor cell-free study on impact of sparge method on dCO_2 removal: (a) 0.033 vvm, (b) 0.0067 vvm, (c) compiled data over a range of vvm. The values in parentheses are the tip speeds for a given condition.

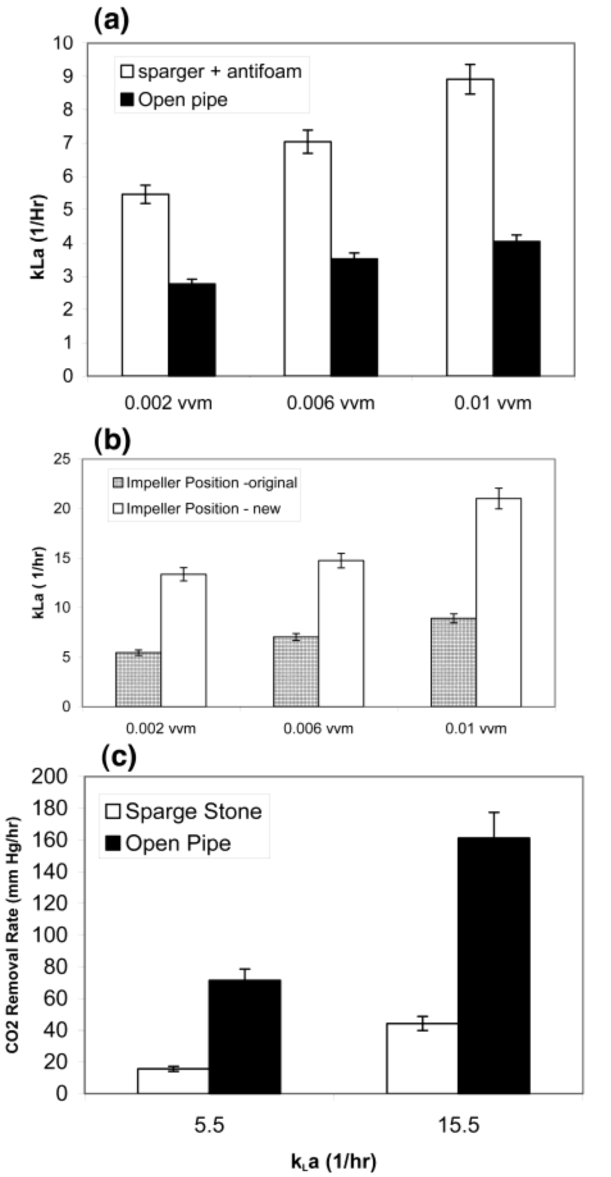


Figure 4. Data from 1000-L cell-free study: (a) impact of sparge rate on k_La, (b) impact of the top impeller position on k_La in the presence of sparge stone, (c) impact of sparge method on dCO_2 removal. Variability estimated from interaction terms for the design of experiment (DOE).

improved k_La for CO_2 and therefore improved CO_2 removal rate. Figure 3c indicates that dCO_2 removal was a strong function of sparge rate in case of spargers and a relatively weak function of sparge rate for open pipe. In a separate study, when 5-fold higher sparge rate was used with an open pipe compared to the sparge rate used for the sparger, the foam level was still almost 10-fold lower (data not shown).

1000-L Cell-Free Study. Increase in sparge rate improved k_La with both sparger and open pipe (Figure 4a). Similar to the 1.5-L cell-free study, k_La was a stronger function of sparge rate in the presence of sparger than in the case of open pipe. k_La was a weak function of tip speed below 2.6 m/s. For open pipe, higher tip speeds were tested and significant improvement in k_La was observed (Table 3). At high tip speed, shear damage is a concern; therefore, feasibility of using high tip speed always has to be tested in cell cultures. For open pipe, foam level at 3.5 m/s tip speed and 0.01 vvm was still lower than the foam level at 1.3 m/s tip speed and 0.002 vvm in the presence of a sparger. The bubbles were

Table 3. Impact of Tip Speed on k_La at 1000-L Scale

	k_La (h^{-1})	
tip speed (m/s)	sparger	open pipe
1.3	7.04	3.44
1.8	7.48	3.79
2.6	na	9.93
3.5	na	15.5

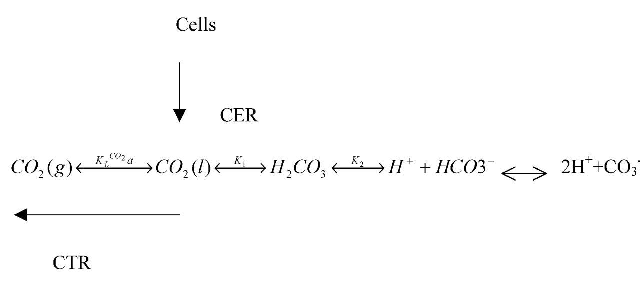
larger, and the foam was in general more stable and tended to stay close to the agitator shaft rather than being equally spread over the liquid surface. Headspace aeration rate and sodium bicarbonate were found to have negligible impact on k_La, whereas antifoam in the case of open pipe was found to decrease k_La by 25%.

The H(top)/D(i) ratio originally was 0.32 compared to the ideal value of 1. The height of the top impeller was decreased, which changed the H(top)/D(i) value to 0.60. This change improved k_La by 2- to 2.5-fold (Figure 4b) in the presence of sparger and by 1.5-fold with an open pipe (data not shown). One qualifier about these data is that these results pertain to a liquid volume of 1000 L and because the working volume changes over the course of the culture due to feeding and sampling of the reactor, the actual k_La values may differ.

For a given k_La value, dCO_2 removal was 3- to 5-fold better with open pipe compared to that with sparger (Figure 4c). dCO_2 removal rate was calculated assuming that dCO_2 decreases linearly with time in the range of 150−100 mmHg. The dCO_2 data acquired in the small-scale study indicated that this was a valid assumption.

Modeling and Small-Scale Study on the Impact of Sodium Bicarbonate Concentration. There are three main sources of dissolved CO_2 in the culture, namely, CO_2 produced by cells during respiration, CO_2 added to culture to maintain pH early in the run, and CO_2 produced from sodium bicarbonate that is added as a buffer at a level of 2 g/L. We wanted to determine whether the reduction in sodium bicarbonate level is a viable option for reducing the dissolved CO_2 level for fed-batch cultures.

Carbon dioxide dissolves in water resulting in the following reactions: where CER = CO_2 evolution rate,



CTR = CO_2 transfer rate, K_1, K_2 = equilibrium constants, $K_L^{CO_2}a$ = mass transfer coefficient

Dissociation of bicarbonate to form carbonate ions is negligible at pH below 7.5 and the concentration of carbonic acid is always very small in comparison to that of dissolved carbon dioxide (*16*). The concentration of bicarbonate ions increases with increasing pH; it is equal to the dissolved CO_2 concentration at a pH of 6.3 and is an order of magnitude greater at pH 7.3. After incorporating the equilibrium between dissolved CO_2 concentration and the partial pressure of CO_2 (pCO_2), the following relationship between bicarbonate concentration and pCO_2 can be established for culture medium at 37 °C and atmospheric pressure (*2*):

$$\log[HCO_3^-] = pH + \log pCO_2 - 7.543 \qquad (1)$$

As Figure 5 shows, for a pH of 6.9, dCO_2 levels in equilibrium with 2, 1, and 0.5 g/L of sodium bicarbonate are 122, 61, and 30.4 mmHg, respectively. In case of perfusion cultures it is well-known that a decreased level of sodium bicarbonate lowers the pCO_2 level throughout the culture time (*3*). This is due to the addition of fresh feed to the reactors throughout culture time. However, in batch and fed-batch cultures the source of sodium bicarbonate is only the initial culture medium. Therefore, the initial benefit of low bicarbonate may or may not continue by the end of the culture. Because cells are a major source of CO_2, to determine the benefits of lower bicarbonate for a fed-batch culture, cellular respiration had to be incorporated. Therefore, a cell culture mathematical model was developed and small-scale (1.5 L) cultures were used instead of a cell-free study. The mathematical model follows a similar rationale as that of Gray et al. for perfusion cultures (*5*).

(i) Model. Dissolved CO_2 and bicarbonate in solution follow eqs 2 and 3, respectively. H_2CO_3 and CO_3^{2-} are present in negligible concentration compared with HCO_3 and CO_2 (*5*):

$$\frac{d[CO_2]}{dt} = CER + k_La([CO_2]^* - [CO_2]) - k_1[CO_2] + k_2[H^+][HCO_3^-] \qquad (2)$$

$$\frac{d[HCO_3^-]}{dt} = k_1[CO_2] - k_2[H^+][HCO_3^-] \qquad (3)$$

Combining eq 2 and 3 gives

$$\frac{d([CO_2] + [HCO_3^-])}{dt} = CER + k_La([CO_2]^* - [CO_2]) \qquad (4)$$

A general form of eq 1 can be used to convert bicarbonate concentration to dissolved CO_2 concentration:

$$\log[HCO_3^-] = pH + \log[CO_2] - pK_1 \qquad (5)$$

Thus eq 4 becomes

$$\frac{d[CO_2]}{dt} = \left(\frac{10^{-pH}}{10^{-pH} + K_1}\right)(CER + k_La([CO_2]^* - [CO_2])) \qquad (6)$$

CO_2 evolution rate (CER) can be described as a function of specific CO_2 evolution rate (SCER, per cell) and cell density:

$$CER = SCER \cdot X_0 \cdot e^{ut} \qquad (7)$$

Equation 6 was solved for a batch process:

$$CO_2 = \frac{A \cdot CER}{\mu + A \cdot k_La} + (\text{constant})(e^{-Ak_Lat}) + CO_2^* \qquad (8)$$

where $$A = \frac{10^{-pH}}{10^{-pH} + K_1} \text{ from eq 6}$$

It should be mentioned that although solutions based on similar sets of variables and assumptions for perfusion cultures have been published, none have been published for batch cultures.

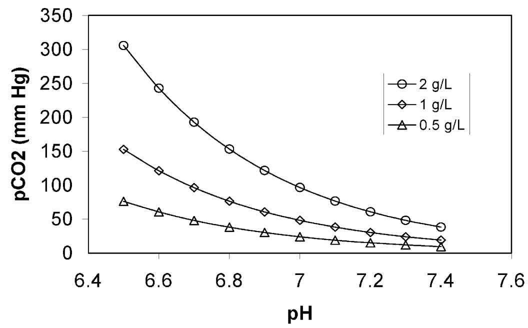


Figure 5. pH, bicarbonate, and pCO2 equilibrium.

Unexpectedly, the results predicted by this model for bicarbonate concentrations of 0.0, 0.5, and 2.0 g/L were identical except for the first couple of hours of culture. What this indicates is that the CO_2 produced by cellular respiration is significantly higher than the dCO_2 generated by sodium bicarbonate in an extended fed-batch culture. The model fit was tested using protein X data from the 1000-L scale (Figure 6a). In this model, the value of $K_1 = 5.2 \times 10^{-7}$ M (*3*) and SCER = 0.35 pmol/cell/h (experimentally calculated oxygen uptake rate, OUR, and assumed that CER/OUR = 1). The model fits reactor data very well in the exponential growth phase of the cells. This is expected as the model uses an exponential term for cell growth. During the first 3−4 days of culture dCO_2 gets added to the culture to maintain pH, as lactic acid production is low. Since the model does not incorporate this factor, the initial part of the model does not represent actual culture condition. All three sodium bicarbonate concentrations generated similar dCO_2 profiles, as indicated by the inset plot in Figure 6a.

If we consider the total CO_2 produced by cells in absence of any CO_2 removal, in a 12-day batch culture the CO_2 level will be 305 mmols/L following the above equation for CER. At pH 6.9, the CO_2 generated by 2 g/L sodium bicarbonate is 4 mmol/L. Therefore, the contribution of sodium bicarbonate to total CO_2 production is minimal (1.3%). Also, even with the original sparging scheme, removal of CO_2 generated by the 2 g/L of sodium bicarbonate will require only 7.8 h, whereas the typical length of a culture is 288 h. Therefore, the time required to remove the CO_2 generated by sodium bicarbonate is only 2.7% of the total culture time. Both of these orthogonal methods indicate that change in sodium bicarbonate level will not have a significant impact on dCO_2 level at the end of culture.

(ii) Small-Scale Cell Culture Data. The purpose of this experiment was to compare the model prediction of impact of bicarbonate with cell culture data. The results from the 1.5-L reactor runs supported the prediction by the model that for batch cultures change in sodium bicarbonate level did not significantly impact the dissolved CO_2 level at the end of the culture (Figure 6b). The cultures containing 2 and 0.5 g/L sodium bicarbonate had similar integrated viable cells (IVC) and titer (data not shown). Since lower sodium bicarbonate level would cause lower buffering capacity and, therefore, maintaining pH in a tight range would be difficult it was decided not to pursue lower sodium bicarbonate level as a strategy for lowering dissolved CO_2 level.

Proof of Concept Runs at 1000-L Scale. The purpose of this study was to incorporate the final CO_2 removal strategies in 1000-L cell cultures to demonstrate

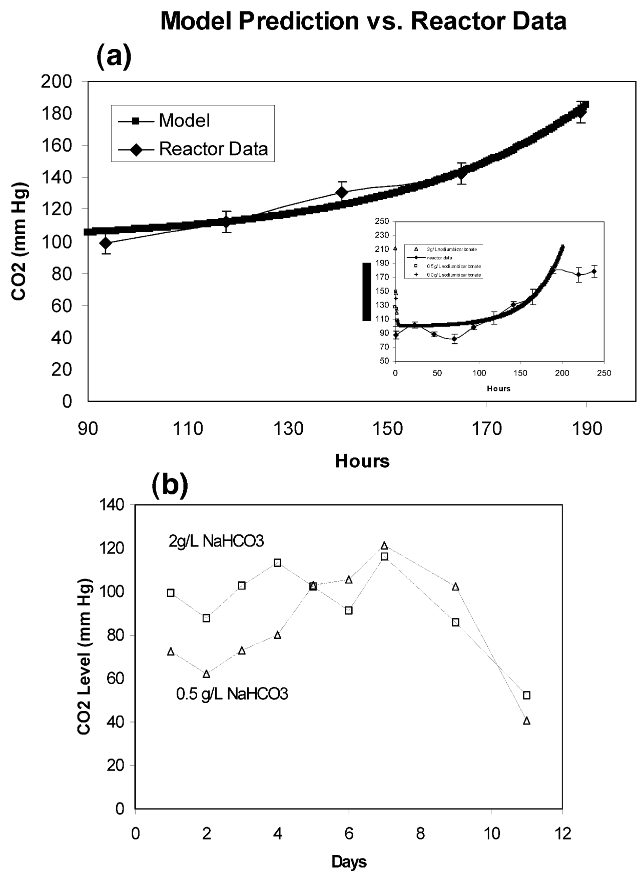


Figure 6. (a) Impact of sodium bicarbonate level on CO_2 at 1.5-L scale (pH = 6.9, temperature = 37 °C), (b) model prediction vs reactor data ($N = 7$) at 1000 L for dCO_2 in fed-batch reactor. For the model, three different sodium bicarbonate concentrations, 2, 0.5, and 0.0 g/L, were tested.

proof of concept. Two 1000-L CHO cultures producing protein X were carried out, one in the presence of a sparge stone and the other with an open pipe. The data generated from the two runs were compared to historical runs. These historical runs resulted from a previous campaign that was conducted to produce protein X for toxicology studies. During the historical runs excessive foam dictated early harvest at day 10. At day 10, dCO_2 level was 179 ± 9 mmHg and further sparging was not possible because of the foam level (Figure 1, dCO_2 at 1000-L data). The goal for the 1000-L runs was to maintain dCO_2 at 70 ± 30 mmHg at the end of culture and to extend culture duration to 14 days, which was the duration at the bench scale.

Final dCO_2 level (with both strategies) was 2.5-fold lower compared to the historical level (72 vs 180 mmHg) (Figure 7a). Total gas sparge for open pipe, antifoam/sparger, and historical data are shown in Figure 7b. The decreased dCO_2 level did not have significant effect on IVC (Figure 7c). However, specific cellular productivity increased by 40% under decreased dCO_2 level (data not shown). Also, since both strategies contributed towards reduction in foam level, it was possible to extend the culture length by 4 days (10 vs 14 days). The increase in culture length resulted in 60% increase in IVC (Figure 7c). As a result of increase in specific productivity and IVC, a total of 2-fold increase in final titer was observed compared to historical control. Harvest samples from two historical runs (one from each reactor) were assayed with the current two lots to ensure that the comparison of titer data was not confounded by assay variability. No titer loss was observed for the historical samples.

Historical data for protein X had shown that extending culture time may cause reduced sialic acid level. Sialic acid molar ratio was found to be similar for the current runs and the historical controls (the normalized molar

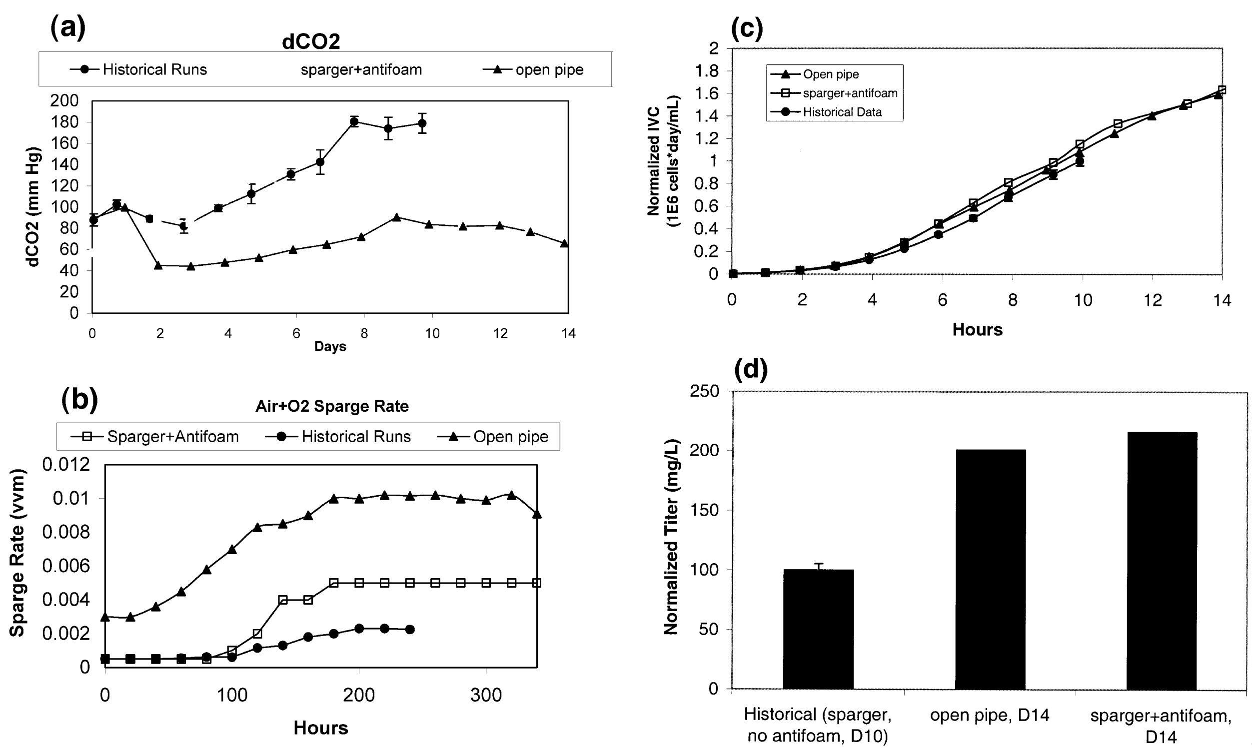


Figure 7. Impact of different dCO_2 removal scheme for 1000-L culture (pH = 6.9, temperature = 37 °C): (a) dCO_2 level, (b) total sparge, (c) integrated viable cell concentration (IVC), and (d) titer at harvest. Normalized titers at day 10 for both of the proof of concept runs were 150 mg/L. Historical run sample numbers are $N = 7$ for a, b and $N = 2$ for c, d.

ratios were 1 and 1.15 for historical runs and the proof of concept runs, respectively).

Conclusions

Our aim was to develop an easily implementable, generic dCO_2 control strategy for large-scale fed-batch processes. Our approach was to first conduct bench-scale (1.5 L) cell-free studies, then conduct large-scale (1000 L) cell free studies, and finally use the best strategy for proof of concept runs at 1000-L scale. In general, the bench-scale and the large-scale cell-free studies showed similar trends. Sparge rate had a strong positive impact on k_La in the presence of a sparger. For open pipe, increased sparge rate had a positive impact also; however, the effect was not as strong. Compared to sparge stone, open pipe generated significantly less foam and for a given k_La also provided better CO_2 removal.

A mathematical model was used to determine the contribution of sodium bicarbonate in the total dCO_2 level at the end of fed-batch culture. The model indicated that cellular respiration was the primary source of dCO_2 for fed-batch culture and reduction in sodium bicarbonate would not have significant impact on the outcome. This prediction was validated in a 1.5-L scale study.

The only difference between the historical and the 1000-L proof of concept runs is the difference in dCO_2 level. The exact same cell line, medium, reactors, and scale-up protocol were used in both cases. Specific metabolite rates for glucose, glutamine, lactate, and ammonia did not change significantly between the historical and present reactor runs. An increase in dCO_2 at constant pH also leads to increased osmolality. Some of the negative impacts observed with high dCO_2 are possibly attributed to an increase in osmolality (*4*, *7*, *14*). In this paper the impact of osmolality has not been decoupled from the impact of dCO_2.

Based on the observations from the 1000L proof of concept runs with open pipe as well as 10-μm sparger (with antifoam), a new protocol for sparging and dCO_2 content has been established. According to this protocol, open pipe with initial addition of 20 ppm of antifoam was used. From an operational perspective, addition of antifoam allows an increase of working volume and reduces the need for visual tank monitoring. Therefore, open pipe with a minimal amount of antifoam was chosen as the final scheme. Use of this protocol for the campaign of a second protein has successfully achieved a dCO_2 level of 61.2 ± 13 mmHg at the end of culture.

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